

**Course: B. Tech.**

**Branch :EXTC**

**Semester : VIII**

**Subject Code & Name: BTETPE 802D Industrial Automation and Control**

**Max Marks: 60**

**Date: 07/07/2022**

**Duration: 3.45 Hr.**

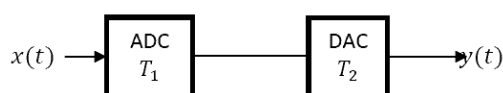
**Instructions to the Students:**

1. All Questions are Compulsory
2. Draw neat diagram wherever necessary.
3. Figures to right indicates full marks
4. Assume suitable data wherever necessary and mention it clearly

(Level/CO) Marks

**Q. 1 Solve Any Two of the following.**

- A) Explain features of IT for the factory that differentiate it with its more ubiquitous counterparts that are used in offices and other business. 06
- B) Consider a simple signal processing system shown in figure below: 06



The sampling periods of the ADC and DAC converters are  $T_1 = 5 \text{ ms}$  and  $T_2 = 2 \text{ ms}$  respectively. Input to the system is given as:  $(t) = 3\cos 100\pi t + 2\sin 250\pi t$ , then output  $(t)$  is

- C) Draw a diagram of Fiber optic position sensor and explain its working. 06

**Q.2 Solve Any Two of the following.**

- A) Discuss Differential Amplifier and derive the equation for same. 06
- B) Discuss with block diagram the Elements of a measuring system. 06
- C) Discuss the values of proportional gain in closed loop continuous oscillation technique. 06

**Q. 3 Solve Any Two of the following.**

- A) Explain the Architecture of PLC with the help of block diagram. 06
- B) Give the significant advantages over conventional control panels of PLC. 06
- C) List and explain Process control input and outputs in stamping process. 06

**Q.4 Solve Any Two of the following.**

- A) Give the Advantages of Hydraulic Actuation Systems. 06
- B) Classify CNC machine tool systems and explain any one way. 06
- C) Dynamic viscosity ( $\mu$ ) of a liquid is  $8.9 \times 10^{-4} \text{ kgm}$ , pipe's cross-section(A) is  $8 \text{ cm}^2$  and hydraulic diameter (DH) of the pipe is  $6 \text{ cm}$ . Reynolds number ( $Re$ ) is  $2.5 \times 10^5$ . Find mass flow rate ( $W$ ) through the pipe. 06

**Q. 5 Solve Any Two of the following.**

- A) Give major aspects of production planning. 06

**B) Explain what is Stator and Rotor in Permanent Magnet Brushless DC Motor.**

**06**

**C) Write a short note on Variable Reluctance type Step Motor and give its specification.**

**06**

**\*\*\* End \*\*\***